

PCA output from FactoMiner

EXHAUSTIVE PCA

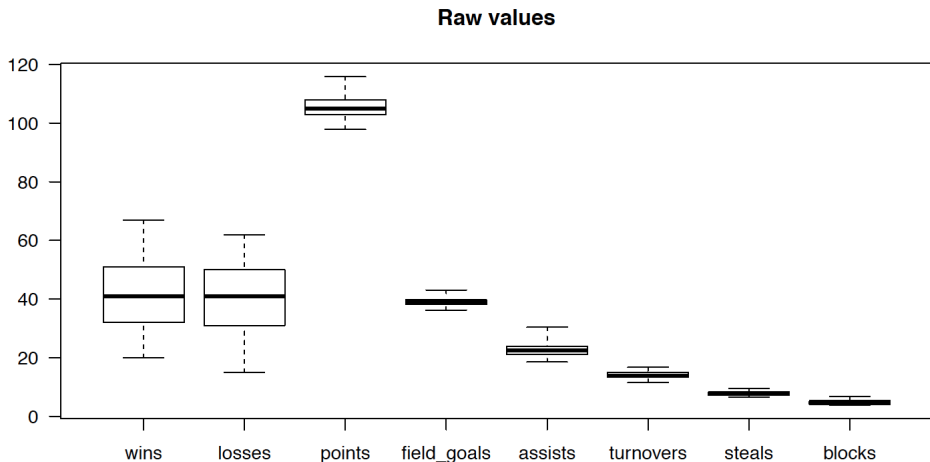
- Data transformations (e.g. usually standardized data).
- Determine active individuals and variables.
- Determine supplementary individuals and variables.
- Interpretation tools: plots, quality measures, contributions, ...

To standardize or not?

- A key issue has to do with the scale of the variables.
- If variables have different units of measurement, then we should standardize them to avoid variables with larger scales dominate the analysis.
- If variables have the same units:
 - you could leave them unstandardized,
 - or you could standardize them (strongly suggested)

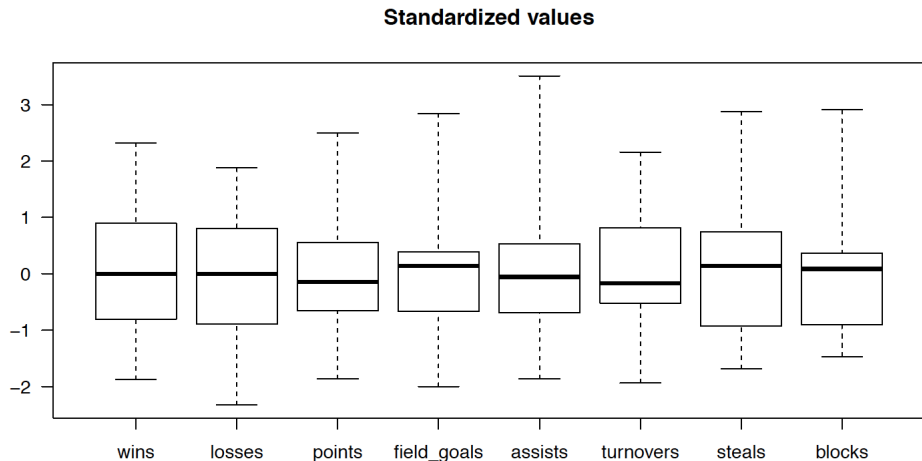
Regardless of the scaling decision, we operate on centered data.

DATA TRANSFORMATIONS



If you use raw scales, wins and losses will dominate the analysis due to their larger variances.

DATA TRANSFORMATIONS



By standardizing the variables, they all play the same role.

ACTIVE VS SUPPLEMENTARY

Active

Selected individuals (rows) and variables (columns) that are used to compute eigenvalues, eigenvectors and PCs.

Supplementary

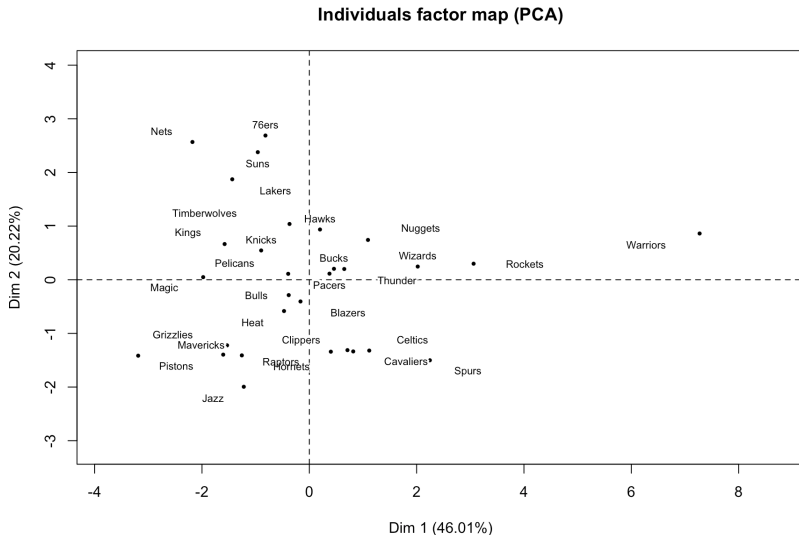
Additional individuals and variables not used in the computational steps, but taken into account for interpretation purposes and/or further data exploration.

STUDYING THE INDIVIDUALS (OBSERVATIONS)

When studying the individuals, we typically pay attention to:

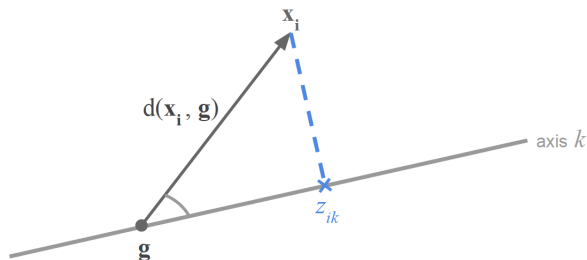
- Scatterplots of PCs
- Quality of representation
- Individual contributions to PCs

SCATTERPLOT OF PCs



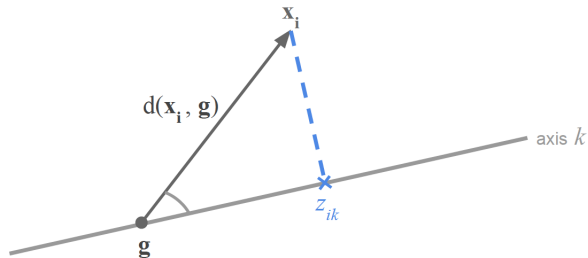
QUALITY OF REPRESENTATION

In addition to the PC plots, you can assess the quality of the individuals representation on a given dimension.



How good is the projection z_{ik} on axis k ?

QUALITY OF REPRESENTATION



$$\cos^2(i, k) = \frac{z_{ik}^2}{d^2(x_i, g)}$$

QUALITY OF REPRESENTATION

$$\cos^2(i, k) = \frac{z_{ik}^2}{d^2(\mathbf{x}_i, \mathbf{g})}$$

where:

- z_{ik} is the value of k -th PC for individual i .
- $d^2(\mathbf{x}_i, \mathbf{g})$ is the squared distance of individual i to the centroid $\mathbf{g}$.
- with centered data $\mathbf{g}$ is the origin.

Cosine² equal to 1 indicates that individual i is exactly on axis k .
Conversely, a cosine² equal to 0 indicates that individual i is on an orthogonal direction of axis k .

QUALITY OF REPRESENTATION

Adding the squared cosines over all principal axes for a given individual, we get:

$$\sum_{k=1}^r \cos^2(i, k) = 1$$

This sum provides, in percentages, the *quality* of the representation of an individual on the subspace defined by the principal axes.

QUALITY OF REPRESENTATION

- The squared cosine is used to evaluate the quality of the representation.
- On a given PC, some distances between individuals will be well represented, while other distances will be highly distorted.
- You can add the squared cosines of an individual over different axes, resulting in a *quality* measure of how well that individual is represented in that subspace.

INDIVIDUAL'S CONTRIBUTION

Another diagnostic tool involves assessing the importance of an observation for a given component by calculating what is known as a **contribution**:

$$ctr(i, k) = \frac{m_i z_{ik}^2}{\lambda_k} \times 100$$

where:

- m_i is the mass or weight of individual i , in this case is $\frac{1}{n-1}$
- z_{ik} is the value of k -th PC for individual i .
- λ_k is the eigenvalue associated to k -th PC.

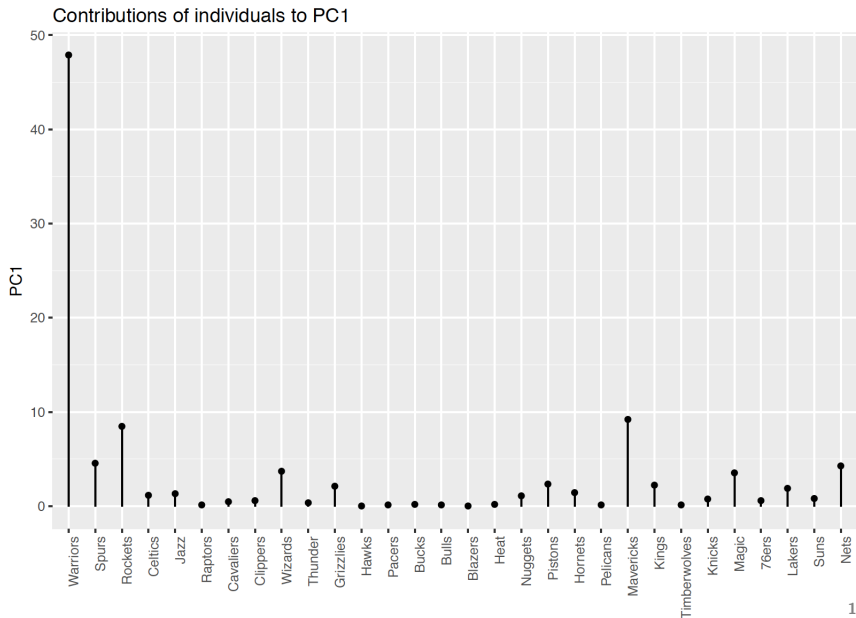
CONTRIBUTIONS

- $ctr(i, k)$ is the contribution of an individual to the construction of component k .
- The contribution is calculated as the percentage of inertia explained by the individual i on component k .
- The value of a contribution is between 0 and 100.
- For a given component, the sum of the contributions of all observations is equal to 100.
- The larger the value of the contribution, the more the individual contributes to the component.

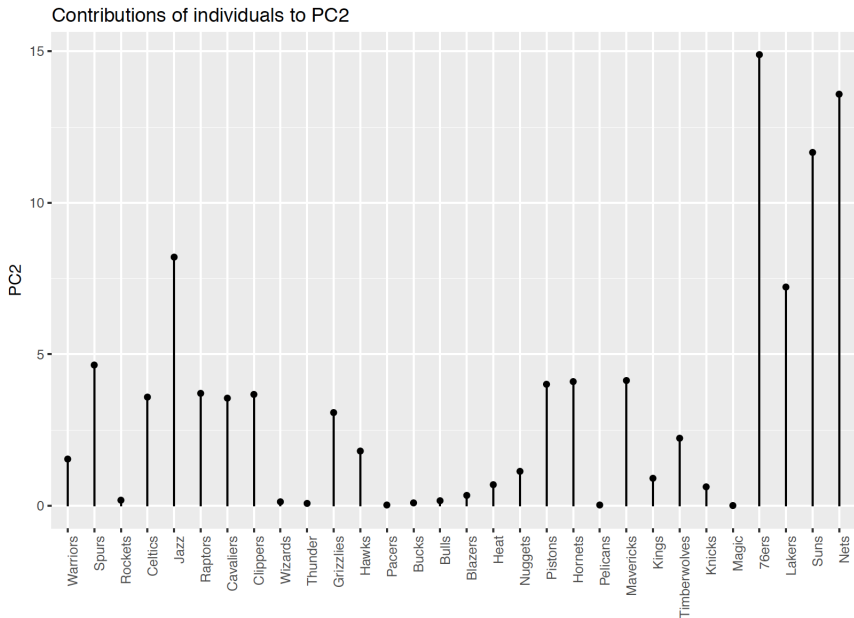
CONTRIBUTIONS

- A useful heuristic is to examine those contributions that are larger than the average contribution: larger than $\frac{100}{(n-1)}$.
- If all individuals had a uniform contribution to a given PC, then they would have a contribution of $ctr(i, k) = \frac{100}{(n-1)}$.
- Outliers have an influence, and it is interesting to know to what extent their influence affects the construction of the components.
- Detecting those individuals that contribute to a given PC helps to assess the component's stability.

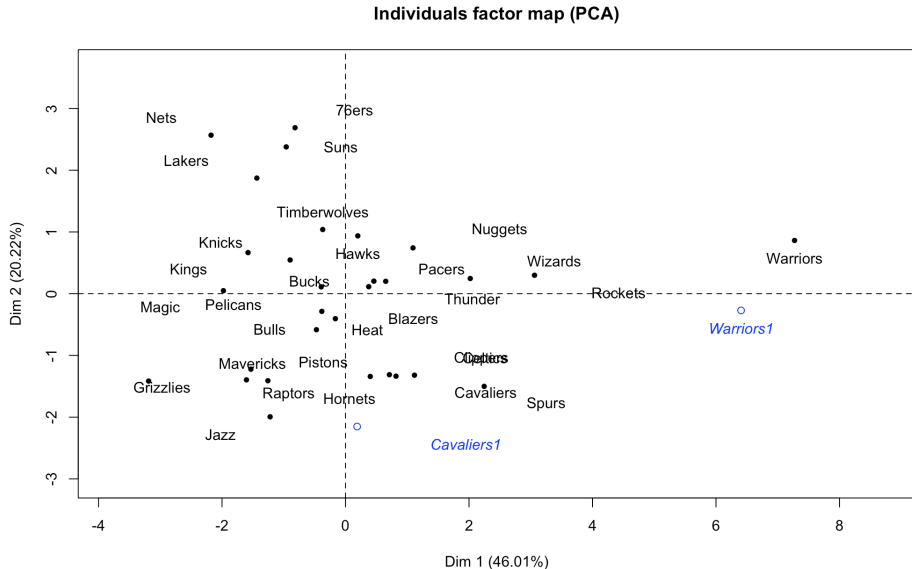
INDIVIDUAL'S CONTRIBUTION



INDIVIDUAL'S CONTRIBUTION



REPRESENTATION OF SUPPLEMENTARY INDIVIDUALS



STUDYING THE VARIABLES

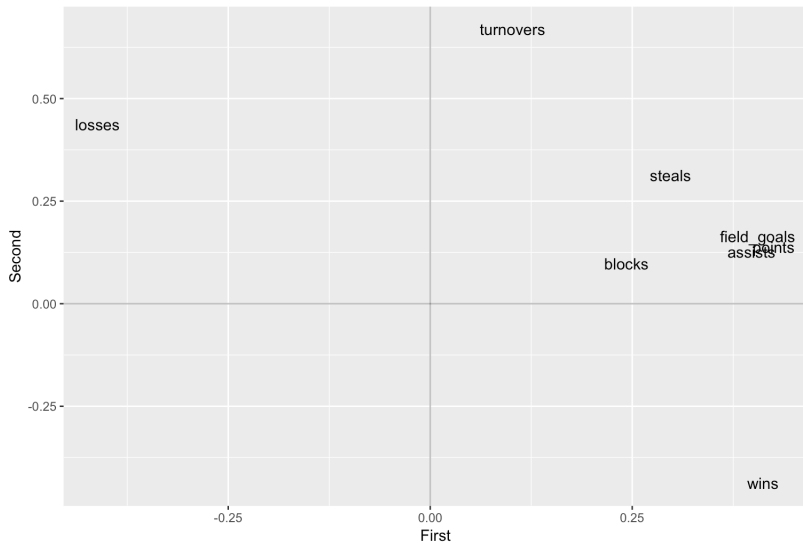
When studying the variables, we typically pay attention to:

- Scatterplots of loadings (or some loading-based result)
- Quality of representation of variables
- Variables contributions to PCs

INTERPRETING PCs

- Being linear combinations of the studied variables, PCs can sometimes be interpreted
- Analysts try to label them in some meaningful way
- This is useful, but not always possible
- You can look at the magnitude of the loadings
- You can also look at the correlations between variables and PCs

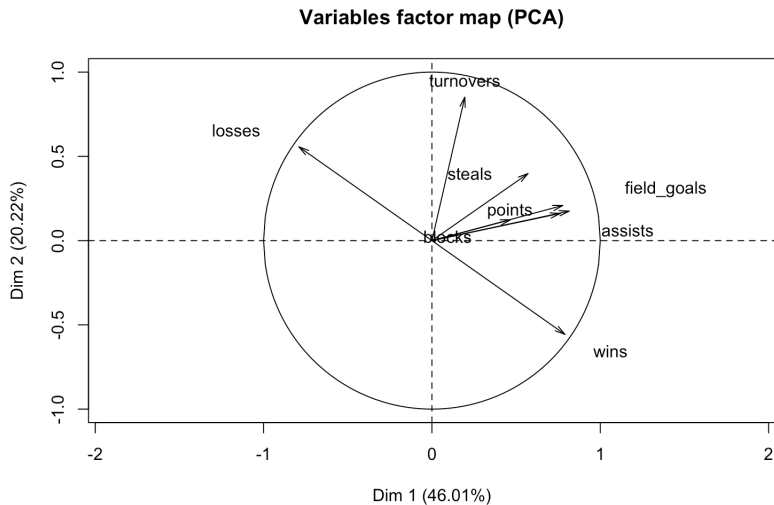
SCATTERPLOT OF VARIABLES (EIGENVECTORS)



CORRELATIONS BETWEEN VARIABLES AND PCs

A more informative interpretation can be obtained by calculating the correlations between the Variables and PCs, and use them to plot a Circle of Correlations. These correlations are collinear with the eigenvectors and in FactoMiner's output are given in `pca_object$var$cor`.

CIRCLE OF CORRELATIONS



SQUARED CORRELATIONS

- The correlation between a component and a variable estimates the information they share.
- Note that the sum of the squared coefficients of correlation between a variable and all the components is equal to 1.
- As a consequence, the squared correlations are easier to interpret than the loadings.
- This is because the squared correlations give the proportion of the variance of the variables explained by the components.
- In FactoMiner's output you can find them in `pca_object$var$cos2`

REPRESENTING SUPPLEMENTARY VARIABLES

Variables factor map (PCA)

